QUASI-REGULAR AND H1-REGULAR SEQUENCES ARE PRESERVED BY BASE CHANGE WHEN THE QUOTIENT IS FLAT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Flatness of the quotients by powers of the ideal follows from flatness of the graded pieces, by induction on the short exact sequences. Algebra, Lemma \ref{algebra-lemma-flat-ses}, cited at proof lines 8008-8009. Gives flatness of the quotient by every power of the ideal.
- Powers of the base changed ideal are the base changes of the powers. Algebra, Lemma \ref{algebra-lemma-flat-tor-zero}, cited at proof lines 8010-8012. Transfers the graded pieces of the ideal to the base changed algebra.
- Vanishing of the first Koszul homology is preserved by base change. Lemma \ref{lemma-base-change-H1-regular}, cited at proof lines 8019-8021. Gives the H1-regular statement.

Source reference. Original lemma 0CEP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15584–15584. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00HM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 648–648. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 647–647. Navigation pointer to a statement the

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Source reference. Original lemma 063Q: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7839–7839. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $A \to B$ and $A \to A'$ be ring maps. Set $B' = B \otimes_A A'$. Let $f_1, \ldots, f_r \in B$. Assume $B/(f_1, \ldots, f_r)B$ is flat over A*

1. *If $f_1, \ldots, f_r$ is a quasi-regular sequence, then the image in B' is a quasi-regular sequence.*
2. *If $f_1, \ldots, f_r$ is a H_1-regular sequence, then the image in B' is a H_1-regular sequence.*

Proof. Assume $f_1, \ldots, f_r$ is quasi-regular. Set $J = (f_1, \ldots, f_r)$. By assumption J^n/J^{n+1} is isomorphic to a direct sum of copies of B/J hence flat over A. By induction and Algebra, Lemma 00HM we conclude that B/J^n is flat over A. The ideal $(J')^n$ is equal to $J^n \otimes_A A'$, see Algebra, Lemma 00HL. Hence $(J')^n/(J')^{n+1} = J^n/J^{n+1} \otimes_A A'$ which clearly implies that $f_1, \ldots, f_r$ is a quasi-regular sequence in B'.

Assume $f_1, \ldots, f_r$ is H_1-regular. By Lemma 063Q the vanishing of the Koszul homology group $H_1(K_\bullet(B, f_1, \ldots, f_r))$ implies the vanishing of $H_1(K_\bullet(B', f_1', \ldots, f_r'))$ and we win. □